

From Implicit Fields to Explicit Primitives: A Synthesis of Architectural Paradigms in Dynamic 3D Gaussian Splatting

The landscape of dynamic scene reconstruction has undergone a paradigmatic shift from computationally intensive implicit neural fields to efficient explicit Gaussian primitives, driven by the demand for real-time rendering in latency-sensitive applications [26] [149]. This survey synthesizes recent advancements in extending 3D Gaussian Splatting (3DGS) to dynamic environments, critically analyzing the trade-offs between reconstruction fidelity, memory footprint, and computational efficiency. We categorize prevailing methodologies into deformation-based paradigms, which warp canonical Gaussians via learned fields [48], and hybrid strategies that disentangle static and dynamic components to minimize redundancy [246]. Furthermore, we examine emerging motion-aware partitioning and trajectory factorization techniques that address the limitations of unified deformation models in capturing complex, non-rigid motions [4] [112]. Despite significant progress, contradictions persist regarding the optimal balance between structure-guided compactness and Gaussian-centric speed, particularly under sparse-view constraints [147]. We conclude by identifying critical gaps in handling irreversible topological changes and integrating physics-informed priors, outlining a roadmap for unified frameworks capable of adaptive resource allocation in unconstrained dynamic settings.

Introduction to Dynamic Scene Representation Paradigms

The landscape of 3D scene reconstruction has undergone a paradigmatic shift, transitioning from implicit neural representations to explicit Gaussian primitives, driven by the critical need for real-time rendering capabilities and high-fidelity synthesis in dynamic environments. While Neural Radiance Fields (NeRF) initially established a robust foundation for photorealistic novel view synthesis by modeling scenes as continuous volumetric functions, the computational intensity associated with ray-marching has historically limited their applicability in latency-sensitive tasks such as robotics and virtual reality [26] [149]. In response, 3DGS emerged as an efficient alternative, utilizing explicit point-based rasterization to achieve unprecedented rendering speeds while maintaining visual quality [18]. However, extending these static representations to dynamic (4D) environments introduces profound challenges related to motion modeling, temporal consistency, and data sparsity, necessitating a diverse array of methodological innovations that balance reconstruction fidelity, memory footprint, and computational efficiency [6] [80].

A dominant strategy in addressing dynamic reconstruction involves the use of deformation fields to warp a canonical set of Gaussians across time, allowing the scene to be represented in a static coordinate system while learning a time-dependent transformation to account for motion. Deformable 3D Gaussian methods learn scenes in a canonical space coupled with a deformation field, achieving real-time rendering speeds superior to implicit dynamic NeRFs [48]. To enhance robustness against sparse initializations and accelerate rendering, other frameworks integrate per-Gaussian embeddings with coarse-to-fine temporal strategies [82]. However, relying on a single global deformation field can lead to blurred renderings in highly dynamic regions due to the inability of a unified model to capture diverse motion patterns [4]. To mitigate this, recent works propose motion-aware partitioning, where Gaussians are recursively split based on dynamic scores, allowing specialized modeling for intricate motions while treating static regions efficiently [4]. Alternatively, some approaches reinterpret 4D Gaussian Splatting as a continuous-time dynamical system, where scene motion arises from integrating a learned neural dynamical field rather than applying per-frame deformations, enabling temporal extrapolation and compositional dynamics [199].

Contrasting with deformation-based paradigms, another significant line of research focuses on hybrid representations and static-dynamic decomposition to optimize memory and computational footprint. The redundancy of modeling static background elements with time-varying parameters has spurred the development of methods that explicitly disentangle these components. Hybrid Gaussian Splatting (HGS) employs a Static-Dynamic Decomposition strategy, utilizing time-dependent Radial Basis Functions for dynamic regions while sharing temporally invariant parameters for static areas, resulting in drastic model size reductions [246]. Similarly, other approaches iteratively convert temporally invariant 4D Gaussians into static 3D representations, significantly reducing parameter counts without sacrificing fidelity in dynamic elements [20]. Furthermore, voxel-based hybrids like 4D Neural Voxel Splatting combine voxel representations with neural Gaussian splatting to model temporal dynamics compactly, avoiding the memory overhead of replicating Gaussians across frames [122]. Empirical investigations suggest that structure-guided methods, which employ auxiliary representations like deformation fields or grids, generally achieve superior reconstruction fidelity and compactness, whereas Gaussian-centric approaches that encode dynamics directly into primitives offer higher rendering speeds but with greater quality variability [147].

Motion modeling techniques have also diversified beyond simple deformation fields to include trajectory factorization and control point-based systems. Recognizing that complex scene motions can often be decomposed into a limited set of basis trajectories, methods like DynMF factorize the motion field into shared neural trajectories, enabling interpretable and efficient dynamic synthesis [112]. Others utilize sparse control points to learn compact transformation bases, which are interpolated to yield the motion field for dense Gaussians, facilitating user-controlled motion editing [223]. In scenarios involving large-scale motions, such as urban driving or sports, relay mechanisms and motion graphs have been introduced to break down complex trajectories into manageable segments, ensuring coherent reconstruction where traditional methods fail [213] [57]. Additionally, physics-informed approaches are beginning to emerge, imposing constitutive laws on Gaussian particles to ensure physically plausible motion patterns in monocular settings [159].

Despite these advancements, significant trade-offs remain between reconstruction quality, rendering speed, and memory efficiency. The challenge of data sparsity, particularly in monocular settings, further complicates these trade-offs, prompting the integration of foundation models and geometric priors to resolve ambiguities in pose and

motion estimation [105]. As the field progresses, the synthesis of these diverse methodologies-ranging from deformation fields and hybrid decompositions to motion factorization and physics constraints-continues to refine the balance required for practical applications in domains as varied as autonomous driving, robotic surgery, and immersive telepresence [3D Gaussian Splatting in Robotics] [80]. Future research must address the lingering gaps in handling irreversible topological changes, ensuring long-term temporal consistency in streaming scenarios, and developing unified frameworks that can adaptively allocate resources based on scene complexity without manual intervention.

Evolution from Implicit to Explicit Dynamic Models

The trajectory of dynamic scene reconstruction has undergone a profound paradigm shift, moving from the computational heaviness of implicit neural fields to the efficiency and controllability of explicit Gaussian primitives. Early approaches to modeling dynamic environments relied heavily on implicit representations, where scene deformations were captured via time-variant Multi-Layer Perceptrons (MLPs) or by warping a canonical space through learned deformation fields. While foundational surveys such as [6] established the theoretical groundwork for handling temporal dynamics, these implicit models frequently struggled with slow inference speeds due to the necessity of volumetric ray marching. Furthermore, they often produced blurred details in regions exhibiting high-frequency motion, limiting their applicability in real-time scenarios. The transition toward explicit representations was driven by the need to decouple geometric fidelity from computational cost, a challenge that implicit methods found difficult to overcome without sacrificing rendering performance.

The introduction of 3DGS marked a critical inflection point, replacing volumetric integration with efficient rasterization-based splatting. However, extending this static framework to dynamic scenes initially presented significant hurdles. Naive extensions often involved predicting per-Gaussian deformations at every timestep, leading to a computational surge that threatened to negate the real-time advantages of the splatting framework. As noted in [147], a fundamental trade-off emerged between structure-guided methods, which utilize auxiliary deformation fields for compactness, and Gaussian-centric approaches that encode dynamics directly into primitives for speed. Initial attempts to bridge this gap, such as [48], proposed learning deformations in a canonical space with an annealing smoothing mechanism to mitigate pose inaccuracies. While effective, these early deformation-based strategies still faced challenges in maintaining geometric coherence and avoiding artifacts during complex non-rigid motions, as highlighted in [165], which emphasized the necessity of integrating explicit 3D geometry features to constrain deformation learning.

To address the limitations of per-Gaussian neural inference, recent research has pivoted toward more sophisticated explicit modeling strategies that treat scene motion as a continuous dynamical system or a factorized set of trajectories. A notable advancement is presented in [199], which reinterprets 4D Gaussian Splatting not as a sequence of discrete deformations but as a continuous-time system governed by a learned neural dynamical field. This formulation allows for sample-efficient learning and temporal extrapolation, capabilities largely absent in traditional deformation-field baselines. Similarly, [84] introduces a Dual-Domain Deformation Model that explicitly captures attribute deformations using polynomial and Fourier series fitting, thereby eliminating the need for heavy implicit neural fields while achieving training speeds significantly faster than per-frame modeling. These approaches underscore a growing consensus that explicit parameterization of motion laws offers superior controllability and temporal consistency compared to black-box implicit predictors.

Further refining the balance between efficiency and fidelity, researchers have explored sparse control mechanisms and motion factorization to reduce redundancy. [223] decomposes motion and appearance by employing sparse control points to learn compact transformation bases, which are then interpolated to drive dense Gaussians. This reduces learning complexity and enhances spatial coherence. In a similar vein, [112] argues that complex scene motions can be decomposed into a small set of learned basis trajectories, enabling real-time rendering speeds exceeding 120 FPS while maintaining interpretability. However, a key open question remains: how to automatically determine the optimal number of basis trajectories or control points for scenes with varying degrees of complexity without manual tuning? Additionally, [14] proposes leveraging optical flow priors to guide these deformation processes, addressing the ambiguity inherent in monocular dynamic reconstruction by establishing robust cross-dimensional motion correspondence.

Despite these advances, challenges regarding memory overhead and large-scale deformation persist, particularly when dealing with non-rigid objects that undergo topological changes. Methods like [90] integrate explicit mesh constraints with Gaussian learning to regularize distributions and enable interactive, large-scale deformations that pure point-based methods struggle to handle. By binding Gaussians to an explicit mesh, this approach suppresses poor-quality primitives and leverages existing mesh deformation datasets for more realistic data-driven results. Meanwhile, [5] addresses the bottleneck of per-Gaussian inference through temporal sensitivity pruning and group-based motion prediction, achieving substantial reductions in model size and rendering latency. The evolution continues with hybrid approaches

such as [246], which explicitly disentangles static and dynamic regions to minimize parameter redundancy, and [25], which utilizes feed-forward encoders to eliminate redundant computations on static regions. Collectively, these developments illustrate a clear trend: the field is moving away from purely implicit, data-hungry deformation fields toward explicit, physics-informed, and structurally aware Gaussian representations that preserve real-time performance without compromising the fidelity of dynamic reconstruction. Future research must now focus on unifying these disparate explicit strategies into a single framework capable of handling both rigid and non-rigid dynamics seamlessly while ensuring robustness in sparse-view and uncalibrated settings.

Core Challenges in Temporal and Spatial Consistency

The extension of 3DGS from static to dynamic environments introduces profound difficulties in maintaining temporal coherence and spatial fidelity, particularly when dealing with monocular inputs or sparse viewpoints. A primary vulnerability lies in the treatment of static regions within dynamic scenes, where naive time-variant modeling leads to significant redundancy and visual instability. Methods that attempt to fit every training view at various time steps often result in heavily redundant Gaussians, causing jittering artifacts in areas that should remain stationary [175]. This redundancy not only degrades rendering speeds but also exacerbates overfitting, as the model struggles to distinguish between genuine motion and noise in static backgrounds. To mitigate this, recent approaches advocate for decomposing scenes into static and dynamic components, utilizing sparse control points or anchor grids to model motion only where necessary [223; 117]. By filtering out anchors corresponding to static areas, these methods reduce the computational burden and prevent the flickering associated with independent Gaussian optimization in non-moving regions [175; 25]. Furthermore, hybrid strategies that assign time-invariant parameters to static regions while reserving time-dependent representations for dynamic elements have demonstrated the ability to drastically reduce model sizes while maintaining real-time performance [246].

The independence of Gaussian primitives often results in a lack of spatial autocorrelation, making optimization unstable under sparse inputs or when modeling complex, non-rigid motions. Simple deformation fields frequently fail to capture high-frequency motion details, leading to geometric inaccuracies and blurred renderings in highly dynamic regions [14; 4]. The under-constrained nature of dynamic reconstruction from 2D observations means that without robust regularization, models are prone to local optima where temporal consistency is not maintained as in the physical world [14]. This issue is particularly acute in monocular settings, where the lack of multi-view constraints makes accurate geometry recovery and temporal coherence inherently ill-posed [190]. Consequently, relying solely on photometric loss is insufficient; methods must incorporate motion priors, such as optical flow or rigid-body constraints, to guide the deformation of Gaussians and ensure physically plausible trajectories [145; 115]. Strategies such as transient-aware deformation auxiliary modules and uncertainty-aware flow augmentation have been introduced to inject motion information and align Gaussian movements with 2D priors, respectively, thereby stabilizing optimization in dim or ambiguous scenes [14].

Contradictions and trade-offs exist regarding the architectural paradigms used to address these challenges. Structure-guided methods, which employ auxiliary representations like deformation fields or canonical spaces, generally achieve superior reconstruction fidelity and compact model sizes but may struggle with rendering speed compared to Gaussian-centric approaches that encode dynamics directly into primitives [147]. While Gaussian-centric methods enable real-time performance, they often suffer from greater quality variability and substantial storage overhead due to the need for per-Gaussian time-varying parameters [147; 5]. Additionally, the low-frequency nature of implicit neural fields used in some deformation models can hinder the representation of complex, rapid motions, leading to ineffective modeling of fine-grained details [237]. To overcome these limitations, researchers have proposed hierarchical motion modeling and multi-scale dynamics mechanisms that factorize complex motion fields into manageable components, thereby alleviating ambiguity and promoting physically plausible dynamics [52; 287].

The challenge of capturing high-frequency motion is further compounded by the difficulty of optimizing deformation fields in dim scenes or under sparse initialization, where models are vulnerable to local optima and color fidelity loss [82]. The inability of simple deformation fields to handle large-scale motions has led to the development of relay mechanisms and trajectory-based parameterizations that break down complex movements into smaller, manageable segments or smooth basis functions [213; 173]. Despite these advances, the tension between computational efficiency, storage costs, and the ability to characterize complex physical motions remains a critical bottleneck. Future research must address the gap between data-driven motion priors and physical plausibility, particularly in scenarios requiring long-term temporal extrapolation or interaction with unseen dynamic objects. The field currently lacks a unified framework that can simultaneously achieve the compactness of structure-guided methods, the speed of Gaussian-centric approaches, and the robustness of physics-informed regularization without relying on extensive pre-computed masks or dense multi-view inputs [161; 184].

Deformation Strategies and Motion Modeling

A central challenge in dynamic 3DGS is the development of motion modeling mechanisms that balance computational efficiency with the fidelity required to capture complex, non-rigid deformations. Early approaches often relied on a single, unified deformation field to warp a canonical set of Gaussians across time. However, consensus indicates that such global models frequently fail to resolve intricate motion patterns in highly dynamic regions while unnecessarily expending resources on static background elements. To address these limitations, recent research has shifted towards hierarchical, clustered, and hybrid modeling techniques that explicitly decouple static and dynamic components or employ sparse control mechanisms to manage deformation complexity.

One prominent strategy involves the explicit decomposition of scenes to optimize resource allocation and improve reconstruction quality. [52] introduces a hierarchical motion modeling strategy that classifies scene elements based on deformation offset statistics and 2D motion flow consistency, allowing for specialized handling of non-rigid motions. Similarly, [4] addresses the blurring issues inherent in unified models by recursively partitioning high-dynamic Gaussians temporally and duplicating their deformation networks for each segment, while treating low-dynamic regions as static to reduce costs. This logic of eliminating redundant computations is further refined in [25], which utilizes a feed-forward encoder and optical flow to achieve state-of-the-art performance by focusing computation only on dynamic regions. Extending this to streaming scenarios, [152] applies rigid deformation only to motion-related Gaussians identified via clustering-based convex hull algorithms, effectively preventing flickering artifacts and improving storage efficiency.

Beyond simple decomposition, researchers have explored sparse control mechanisms to manage complex deformations more effectively without the overhead of per-Gaussian networks. [223] decomposes motion and appearance by using sparse control points to learn compact 6-DoF transformation bases, which are interpolated to yield the motion field of dense Gaussians, thereby reducing learning complexity and enhancing temporal coherence. [117] proposes a similar sparse anchor-driven framework where a hierarchical propagation mechanism increases anchor resolution based on motion complexity, suppressing redundant updates in static areas. This concept of aligning control density with motion complexity is advanced in [173], which leverages semantic priors to replace MLP-based fields with spline-based trajectory parameterization for smoother motion representation. Furthermore, [1] clusters Gaussians with similar properties into superpoints, treating them as rigid units to accelerate rendering without sacrificing quality, while [5] introduces GroupFlow to cluster Gaussians by trajectory similarity, predicting a single rigid transformation per group to drastically reduce inference costs.

The integration of physical laws and explicit motion priors represents another critical direction, moving beyond purely data-driven deformation fields. [143] formulates each 3D point as a rigid particle, explicitly estimating physical parameters to govern motion without human labels. [158] adopts a similar physics-informed approach but introduces a divergence-free module to estimate per-Gaussian velocity fields, avoiding inefficient PINN losses. [199] reinterprets scene motion as a continuous-time dynamical system integrated from a learned neural field, enabling temporal extrapolation beyond observed ranges. To address the ambiguity of monocular settings, [145] decouples optical flow into camera and motion flows to explicitly constrain Gaussian deformation, while [264] targets motion blur by reparameterizing Gaussian shapes along motion trajectories using kinematic priors.

Addressing the challenges of large-scale and long-duration sequences, several methods employ trajectory-based or segmented modeling to avoid per-frame training overhead. [84] utilizes a Dual-Domain Deformation Model that captures time-dependent residuals via polynomial and Fourier series fitting, eliminating the need for per-frame training. [213] breaks down large-scale motions by replicating foreground Gaussians across temporal segments, acting as explicit relay nodes to simplify complex trajectories. Conversely, [106] employs cluster-centric basis transformations that adaptively expand to capture evolving motion patterns over extended sequences. Finally, novel representations such as forward warping offer alternatives to traditional backward deformation fields; [226] argues that backward flow fields are often discontinuous and proposes estimating a smooth forward flow field to warp canonical radiance fields directly, while [VeGaS: Decoupling Motion and Geometry in 4D Gaussian Splatting] introduces a Galilean shearing matrix to explicitly incorporate time-varying velocity, decoupling motion from geometric covariance to model complex non-linear motions. These diverse strategies collectively demonstrate a shift towards more structured, physically grounded, and computationally efficient motion modeling paradigms, though open questions remain regarding the

generalization of physical priors to unseen object categories and the robustness of sparse control in highly occluded monocular settings.

Canonical Space and Deformation Fields

The predominant paradigm for reconstructing dynamic scenes using 3DGS involves establishing a canonical set of 3D Gaussians and predicting their temporal evolution through learned deformation fields. This approach effectively decouples static geometry from dynamic motion, allowing the network to focus exclusively on necessary transformations rather than relearning the entire scene structure for every frame. A foundational implementation of this strategy is presented in [48], which reconstructs scenes in a canonical space while utilizing a deformation field to model monocular dynamic scenes. However, standard deformation fields that predict transformations for every individual Gaussian can be computationally expensive and prone to blurring, particularly in scenes with complex motion patterns where the degrees of freedom are excessively high. To address geometric incoherence often found in implicit methods, [165] argues for explicitly extracting 3D geometry features to guide the deformation of canonical Gaussians, ensuring that the learned motion remains geometrically consistent.

To mitigate the computational bottlenecks and memory overhead associated with per-Gaussian inference, researchers have introduced clustering mechanisms known as "superpoints" to reduce the dimensionality of the motion representation. The core insight behind [1] is that neighboring Gaussians often share similar rigid motions. By clustering these Gaussians into superpoints based on properties such as rotation, translation, and location, the method significantly reduces the degrees of freedom required to model the scene. This allows the system to compute a single deformation for an entire cluster rather than for every primitive, accelerating inference speeds without sacrificing visual fidelity. This concept of motion grouping is further explored in [5], which introduces "GroupFlow," a technique that clusters Gaussians by trajectory similarity to predict a single rigid transformation per group. Additionally, [4] addresses the limitations of unified models by partitioning Gaussians based on dynamic scores, applying specialized deformation networks only to high-dynamic regions while treating low-dynamic areas as static to reduce costs. These partitioning strategies collectively demonstrate that adaptive resource allocation is critical for scaling dynamic reconstruction to complex, real-world environments.

Hybrid frameworks have also emerged to address the linear memory growth associated with storing per-frame Gaussians or dense deformation fields, moving towards more compact spatiotemporal encodings. [84] proposes a Dual-Domain Deformation Model that captures time-dependent residuals using polynomial fitting in the time domain and Fourier series in the frequency domain. This explicit modeling eliminates the need for separate 3D Gaussians per frame or additional implicit neural fields, achieving training speeds five times faster than per-frame modeling. In a similar vein, [161] introduces a deformable anchor grid where each anchor point derives multiple Gaussians in its local spatiotemporal region, serving as a memory-efficient geometric backbone. Complementary to this, [101] retains a sufficient quantity of Gaussians to model dynamic content accurately while compressing them into compact, grid-aligned 4D anchor features, dynamically growing anchors in under-reconstructed regions. These methods collectively aim to compactly predict deformations, avoiding the storage explosion inherent in naive 4D extensions while maintaining the ability to capture high-frequency temporal details.

Despite these advancements, contradictions and trade-offs remain regarding the expressiveness of deformation models versus their efficiency. While clustering and anchoring improve speed, they may struggle with highly non-rigid or complex motions where individual primitives move independently. To address this, [60] proposes VeGaS, which decouples Gaussian motion from geometric attributes using a Galilean shearing matrix to model complex non-linear motions that rigid clustering might miss. Furthermore, [199] reinterprets the problem entirely, treating the scene as a continuous-time dynamical system where motion arises from integrating a learned neural field rather than applying discrete per-frame deformations. This approach enables temporal extrapolation and compositional dynamics, capabilities often absent in standard deformation-field baselines. Alternative strategies involve leveraging explicit structural priors to guide deformation, such as [90], which binds 3D Gaussians to an explicit mesh to regularize distribution and guide large-scale deformations, thereby preventing artifacts common in purely point-based methods. Similarly, [7] constrains Gaussians to roam near a mesh, creating a mutually adsorbed representation that harnesses both rendering flexibility and structured motion principles. However, reliance on specific priors can limit generalizability; for instance, [264] introduces kinematic priors specifically to handle motion-induced blur, reparameterizing Gaussian shapes along motion trajectories to prevent shape collapse in scenarios where standard deformation fields fail. The tension between reconstruction quality, compactness, and rendering speed remains a central

theme, with recent works like [112] attempting to bridge this gap by factorizing the dense motion field into a tiny set of learned basis trajectories, allowing for real-time rendering while maintaining high photorealistic quality.

Motion-Aware Partitioning and Clustering

The extension of 3DGS to dynamic scenes initially relied on applying a unified deformation field to all Gaussians, a strategy that often resulted in blurred renderings and a loss of fine motion details in highly active regions due to the inability of a single model to capture diverse motion patterns [4]. To mitigate these inefficiencies, recent research has shifted towards motion-aware partitioning and clustering strategies that adapt the complexity of the motion representation to the actual dynamics of the scene. A primary approach involves distinguishing between static and dynamic components to reduce computational redundancy. For instance, MAPo introduces a dynamic score-based partitioning strategy that recursively partitions high-dynamic Gaussians temporally while treating low-dynamic regions as static, utilizing specialized deformation networks for complex segments to preserve intricate details [4]. This separation is further refined in frameworks like MGStream, which employs motion masks and clustering-based convex hull algorithms to isolate motion-related Gaussians, applying rigid deformation only where necessary to avoid flickering artifacts and improve storage efficiency [152]. Similarly, DynaSplat classifies scene elements through a fusion of deformation offset statistics and 2D motion flow consistency, enabling a hierarchical motion modeling strategy that captures both coarse global transformations and fine-grained local movements [52].

Complementing the static-dynamic dichotomy, trajectory clustering has emerged as a powerful technique to group Gaussians with similar motion characteristics, thereby reducing the dimensionality of the deformation problem. Speedy Deformable 3D Gaussian Splatting introduces GroupFlow, a motion analysis technique that clusters Gaussians by trajectory similarity and predicts a single rigid transformation per group rather than separate deformations for each Gaussian, significantly pruning redundant computations and accelerating rendering speeds [5]. This concept of grouping is echoed in Superpoint Gaussian Splatting, which clusters Gaussians with similar properties such as rotation and translation into "superpoints," allowing the system to compute deformations for the group rather than individual primitives, thus extending 3DGS to dynamic scenes with minimal computational overhead [1]. Furthermore, SC-GS decomposes motion and appearance by using sparse control points to learn compact 6-DoF transformation bases, which are interpolated to yield the motion field of dense Gaussians, effectively reducing learning complexities and enhancing temporal coherence [223]. The efficacy of clustering is also demonstrated in DynMF, which factorizes the dense motion field of a scene into a small set of learned neural trajectories, binding each point to motion coefficients that enforce the sharing of basis trajectories, thereby achieving real-time rendering speeds comparable to static 3DGS [112].

While many methods focus on grouping or partitioning, others explore hierarchical or anchor-based structures to manage motion complexity. SD-GS proposes a deformable anchor grid where each anchor point derives multiple 3D Gaussians, employing a deformation-aware densification strategy that adaptively grows anchors in under-reconstructed high-dynamic regions while reducing redundancy in static areas [161]. This aligns with HAIF-GS, which utilizes an Anchor Filter to suppress redundant updates in static areas and a Hierarchical Anchor Propagation mechanism to increase anchor resolution based on motion complexity, ensuring structured and consistent dynamic modeling [117]. In contrast to these explicit grouping methods, some approaches leverage physical priors to guide motion modeling. TRACE formulates each 3D point as a rigid particle, learning a translation-rotation dynamics system that allows objects to be segmented simply by clustering learned physical parameters, offering a distinct perspective on motion decomposition compared to purely geometric clustering [143]. Similarly, NeHaD incorporates Hamiltonian mechanics to govern deformation fields, using Boltzmann equilibrium decomposition to adaptively separate static and dynamic Gaussians based on their spatial-temporal energy states, providing a physics-informed alternative to heuristic partitioning [46].

Despite the consensus on the benefits of partitioning and clustering, contradictions exist regarding the optimal granularity and representation of these groups. While methods like Speedy Deformable 3DGS and Superpoint Gaussian Splatting advocate for rigid transformations within clusters to maximize speed, others like MAPo and DynaSplat argue for more flexible, non-rigid modeling in high-dynamic regions to prevent visual discontinuities and capture complex deformations [4] [52]. Additionally, the reliance on optical flow or motion masks for initial partitioning, as seen in MotionGS and MGStream, introduces a dependency on the accuracy of these auxiliary inputs, which can be problematic in monocular settings with ambiguous motion cues [145] [152]. Conversely, methods like DynMF and TRACE attempt to learn motion structures directly from data without heavy reliance on external flow supervision, though this may require more complex optimization strategies. The trade-off between the compactness of clustered

representations and the fidelity of per-Gaussian deformation remains a central theme, with recent works like HGS and Hybrid 3D-4D Gaussian Splatting attempting to balance these by explicitly disentangling static and dynamic regions within a unified representation to minimize parameter redundancy while maintaining high-frequency detail [246] [20]. These diverse strategies collectively highlight a paradigm shift towards adaptive modeling, where the computational cost scales dynamically with scene activity, ensuring efficient rendering without compromising the reconstruction of rapid or complex motions. Future research must address the challenge of determining the optimal clustering granularity automatically and developing robust partitioning mechanisms that function reliably in the absence of accurate external motion priors.

Leveraging Optical Flow and External Priors

Recognizing that image supervision alone is often insufficient for robust motion estimation in dynamic 3DGS, recent research has increasingly integrated external motion priors, particularly optical flow, to guide Gaussian deformation. The fundamental challenge in monocular dynamic reconstruction lies in the ill-posed nature of inferring 3D motion from 2D observations, where ambiguities frequently lead to temporal incoherence and geometric artifacts. To mitigate these issues, motion-aware frameworks have been developed to establish explicit correspondences between 3D Gaussian movements and pixel-level flow fields. For instance, [14] proposes a framework that mines motion cues from optical flow to enhance dynamic modeling, introducing an uncertainty-aware flow loss to align 3D trajectories with 2D observations more robustly than render-based supervision alone. Similarly, [145] introduces an optical flow decoupling module that separates camera flow from object motion flow, using the latter to explicitly constrain the deformation of 3D Gaussians and simulate dynamic object motion accurately. These methods demonstrate that direct flow supervision can significantly reduce optimization ambiguity, though they rely heavily on the accuracy of pre-computed flow estimates, which can introduce noise into the reconstruction process.

Beyond direct flow supervision, some methods leverage pre-trained scene flow fields to fundamentally alter the rendering pipeline rather than just warping geometry. [22] presents a novel approach where a per-scene optimized scene flow field is used to bend camera rays, effectively counteracting scene dynamics. This technique presents the dynamic scene as static to a pre-trained rendering network, drastically reducing optimization time by an order of magnitude while maintaining high rendering quality. This strategy contrasts with traditional deformation fields that warp the scene geometry itself, offering a computationally efficient alternative for handling complex motions. Furthermore, [70] mathematically derives dynamic Gaussian motion directly from optical flow and camera motion constraints, enabling annotation-free optimization and ensuring the continuity of dynamic motions without relying on complex control signal annotations. By treating flow as a derivative of the Gaussian trajectory, this approach bridges the gap between 2D observations and 3D control in a physically consistent manner.

However, significant contradictions exist regarding the dependency on explicit flow labels versus self-supervised induction. While [145] and [14] rely on explicit optical flow inputs, other works argue for unsupervised approaches to avoid the inaccuracies inherent in off-the-shelf flow estimators. [117] employs a self-supervised Induced Flow-Guided Deformation module that induces anchor motion using multiframe feature aggregation, eliminating the need for explicit flow labels while handling fine-grained deformations through hierarchical anchor propagation. Similarly, [243] tackles the ambiguity of lifting 2D flow to 3D by decoupling object and camera motion through unsupervised surface consistency and patch-based multi-view constraints, achieving accurate scene flows without extra supervision. This divergence highlights a critical open question: whether the noise introduced by imperfect external flow estimators outweighs the benefits of explicit guidance compared to learning motion representations directly from multi-view consistency. Recent efforts like [131] attempt to bridge this gap by integrating time-dependent 4D Gaussian representation within Neural Motion Flow Fields to learn space-time representations without tracked 3D bounding boxes, demonstrating that self-supervised flow modeling can achieve state-of-the-art performance in complex urban scenarios.

The integration of external priors also extends to handling specific challenges such as motion blur and sparse viewpoints, where standard photometric losses often fail. [264] addresses the entanglement of object motion and geometry caused by motion-induced blur by introducing a kinematic prior that reparameterizes Gaussian shapes along motion trajectories. This approach mitigates degenerate shape collapse without requiring auxiliary motion supervision, effectively using the blur itself as a cue for motion direction. In scenarios with limited data, [232] enhances the quality of 2D priors from foundation models, combining video segmentation with epipolar-error maps derived from optical flow to generate reliable object-level masks and 2D tracks. These refined priors are then embedded into the reconstruction stage via scaffold-projection and virtual-view depth losses, significantly improving the fidelity of thin structures and complex motions that are often lost in purely optimization-based approaches. Additionally, [287] factorizes complex motion fields into multi-level compositions, incorporating multi-modal priors from vision foundation models to constrain the solution space and promote physically plausible dynamics.

Hierarchical and multi-scale approaches further refine the utilization of flow priors to manage motion complexity across different temporal resolutions. [287] and [117] both highlight a consensus that decomposing motion into manageable

components, guided by external flow information or learned inductions, is crucial for resolving ambiguities in under-constrained scenarios. The efficacy of these flow-guided approaches is also evident in streaming and real-time applications. [152] employs motion-related 3D Gaussians implemented according to motion masks and clustering algorithms to avoid flickering artifacts and improve storage efficiency in streaming contexts. By applying rigid deformation only to motion-related Gaussians, the method ensures temporal consistency while maintaining real-time rendering capabilities. Furthermore, [45] leverages flow priors to enhance temporal high-frequency emphasis reconstruction, focusing optimization on motion-intensive parts to improve dynamic awareness in neural rendering for surgical scenes. These diverse strategies collectively demonstrate that leveraging optical flow and external priors is not merely an auxiliary step but a foundational component for achieving robust, temporally coherent, and high-fidelity dynamic scene reconstruction in 3DGS. Future research must address the reliability of these priors in extreme conditions and explore unified frameworks that can adaptively switch between explicit and implicit motion modeling based on scene complexity.

Enhancing Representational Capacity and Regularization

While 3DGS has revolutionized real-time novel view synthesis through its explicit and efficient rasterization pipeline, the foundational formulation relies on primitives with inherently limited expressiveness. The standard assumption of Gaussian kernels parameterized by spherical harmonics for view-dependent color often struggles to capture high-frequency details, complex specularities, and intricate geometric variations, particularly in sparse-view settings or dynamic environments where data constraints are severe. To address these limitations, a significant body of research has emerged focused on augmenting Gaussian primitives with neural fields, texture maps, and implicit regularizations. These enhancements aim to inject spatial bias and global consistency into the otherwise independent Gaussian primitives, effectively preventing overfitting and improving generalization across diverse tasks. The convergence of explicit Gaussian primitives with implicit neural regularizers represents a strategic synthesis, combining the rendering speed of explicit models with the generalization capabilities of implicit fields.

A primary strategy for enhancing representational capacity involves integrating neural textures or implicit fields directly with the Gaussian primitives to overcome the locality constraints of standard splats. [3] introduces a framework that augments each Gaussian with local RGBA texture fields predicted by a global tri-plane neural network. This approach moves beyond the static color and opacity of standard 3DGS, enabling the modeling of expressive view- and time-dependent effects that are critical for capturing specular reflections and dynamic appearance changes. By leveraging a shared global representation, this method ensures robust spatial consistency across primitives, mitigating the redundancy and overfitting issues often observed in methods that optimize per-splat textures independently. Similarly, [2] addresses the vulnerability of independent splats in sparse-view scenarios by regularizing splat features through an implicit neural field. This introduces a necessary spatial autocorrelation bias, where nearby primitives exhibit similar features, thereby stabilizing optimization when input views are limited. These methods collectively demonstrate that injecting implicit priors into explicit primitives is essential for robust reconstruction under data-scarce conditions, effectively bridging the gap between the compactness of neural fields and the efficiency of splatting.

In the domain of surface reconstruction, the lack of explicit geometric constraints in vanilla 3DGS often leads to noisy or incomplete surfaces, prompting researchers to couple Gaussian splatting with neural implicit surfaces or meshes to enforce geometric fidelity. [179] proposes a joint optimization pipeline where 3DGS provides dense point cloud guidance to a neural implicit surface model. Crucially, this work introduces a scale regularizer to force Gaussian centers to align closely with the underlying surface, refining the point cloud iteratively using normal priors from the neural model. This bidirectional interaction allows the system to recover highly detailed surfaces that neither method could achieve in isolation. Further advancing this hybridization, [81] binds 3D Gaussians directly to mesh faces, enabling end-to-end learning of both explicit geometry and appearance. This mesh-guided approach creates an effective information pathway for photometric supervision, improving efficiency and enabling direct manipulation of the reconstructed scene. Additionally, [90] extends this concept by binding Gaussians to an explicit mesh to facilitate large-scale deformations, leveraging the mesh's topology to guide Gaussian splitting and prevent artifacts during manipulation. Despite these advances, a key open question remains: how to balance the computational overhead of joint optimization with the need for real-time performance in interactive applications without sacrificing geometric accuracy.

The challenge of representational capacity is further compounded in dynamic scenes, where the trade-off between modeling complex motion and maintaining computational efficiency becomes critical. Standard approaches that assign independent deformation fields to every Gaussian often suffer from excessive memory usage and temporal inconsistency. [112] proposes a compact representation that decomposes the complex motion field of a scene into a small set of learned neural trajectories. By binding each Gaussian to a linear combination of these basis trajectories, the method significantly reduces parameter redundancy while effectively disentangling overlapping motions. Complementing this, [1] clusters Gaussians with similar motion properties into "superpoints," allowing for shared deformation calculations that accelerate rendering without sacrificing visual fidelity. For scenarios requiring even greater compactness, [20] adaptively represents static regions with time-invariant 3D Gaussians while reserving full 4D representations only for dynamic elements, drastically reducing the computational load. However, the reliance on clustering or factorization raises concerns about the ability to capture fine-grained, non-rigid deformations that do not adhere to global motion bases, suggesting a need for adaptive mechanisms that can switch between local and global motion models dynamically.

Furthermore, specific applications demand specialized augmentations to handle unique physical phenomena or environmental constraints. For instance, [279] extends the representational capacity of Gaussians to model translucent materials by introducing interior Gaussians and a Fresnel-based blending mechanism, addressing optical properties that standard opaque models cannot capture. In underwater environments, [196] integrates a learnable image formation module to simulate light attenuation and scattering, effectively separating media effects from intrinsic scene radiance. These specialized enhancements demonstrate that augmenting Gaussian primitives with domain-specific neural modules or physical priors is essential for extending 3DGS beyond idealized static scenes into complex, real-world applications. The collective evolution of these methods underscores a clear trend: the future of high-fidelity 3D reconstruction lies not in choosing between explicit or implicit representations, but in their strategic integration to maximize both expressiveness and regularization, while addressing the unresolved challenge of scalable, generalizable learning across diverse physical domains.

Neural Fields and Implicit Regularization

The transition from purely explicit Gaussian primitives to hybrid architectures incorporating implicit neural fields represents a critical evolution in addressing the inherent limitations of standard 3DGS, particularly regarding representational capacity and optimization stability. While the original 3DGS formulation offers unparalleled rendering efficiency through rasterization, it relies on the independent optimization of millions of primitives, each parameterized by position, covariance, opacity, and spherical harmonics. This independence, while computationally advantageous, introduces significant challenges in sparse-view scenarios and complex dynamic environments where data constraints are insufficient to uniquely determine the attributes of every primitive. To mitigate the overfitting and spatial inconsistency common in independent splat optimization, recent methodologies have increasingly turned to implicit neural fields to enforce spatial autocorrelation and introduce continuous biases that regularize the learning process.

A primary strategy for stabilizing optimization involves modeling splat features as outputs of an implicit neural field rather than optimizing them as independent variables. [2] explicitly identifies the lack of spatial autocorrelation among splat features as a fundamental cause of suboptimal performance in sparse reconstruction settings. By regressing splat features through a hierarchical convolutional decoder that outputs a tri-plane representation, this approach enforces a spatial bias wherein nearby primitives are compelled to share similar features. This mechanism effectively mimics the spectral bias inherent in Multi-Layer Perceptrons (MLPs) used in NeRF, ensuring that the optimization landscape is smoother and less prone to converging to local minima driven by noise in limited input views. The result is a consistent enhancement in reconstruction quality for both static and dynamic scenes under unconstrained sparse view conditions, bridging the gap between the robustness of volumetric rendering and the efficiency of splatting.

Expanding upon the concept of hybrid regularization, [3] proposes a unified framework that augments each Gaussian primitive with local RGBA texture fields predicted by a global tri-plane network. Unlike methods that simply attach static texture maps to primitives, which often fail to capture view-dependent effects or generalize to sparse inputs, this approach leverages a shared global representation to model local texture fields across all primitives. This design not only reduces model size and facilitates efficient global information exchange but also enables the modeling of expressive view- and time-dependent effects that standard spherical harmonics cannot adequately capture. The implicit regularization provided by the global neural field ensures robustness across diverse tasks, including geometry reconstruction and dynamic scene modeling, where independent per-splat optimization typically leads to redundant capacity and poor spatial consistency. By conditioning the local texture fields on viewing direction and timestep, the method achieves a level of expressiveness previously reserved for purely implicit models while retaining the real-time rendering capabilities of explicit splatting.

The integration of neural fields also addresses specific limitations in dynamic scene reconstruction, where the sheer number of parameters required to model temporal deformations can lead to excessive memory consumption and training instability. [30] replaces traditional spherical harmonics with neural features splatted from spacetime Gaussians, facilitating the modeling of view- and time-dependent appearance within a compact storage footprint. Similarly, [112] decomposes complex motion fields into a small set of learned neural trajectories, binding each point to motion coefficients that enforce the sharing of basis trajectories. This factorization acts as a powerful regularizer, constraining the inherently under-constrained motion field of a dynamic scene and enabling effective optimization even with monocular inputs. These approaches demonstrate that neural fields can serve not only as regularizers for spatial attributes but also as compact encoders for temporal dynamics, significantly reducing the degrees of freedom required to represent complex 4D scenes.

Furthermore, the synergy between explicit primitives and implicit fields extends to surface reconstruction and geometric fidelity. [179] illustrates a bidirectional benefit where 3DGS provides dense point clouds to guide neural implicit surface reconstruction, while the implicit model refines the Gaussian distribution through normal priors. This mutual reinforcement highlights a broader trend where hybrid models leverage the strengths of both paradigms: the explicit nature of Gaussians for efficient rendering and dense geometry sampling, and the continuous, regularized nature of neural fields for high-fidelity surface definition and noise suppression. Even in specialized domains such as underwater reconstruction, [196] employs voxel-based regression modules that function similarly to neural fields to model spatially varying attenuation and backscatter, demonstrating the versatility of implicit components in handling physical phenomena that pure explicit models struggle to represent.

Despite the clear advantages, the integration of neural fields introduces computational overhead during training and, in some cases, inference. Methods like [184] address this by employing grid-based neural fields to replace spherical harmonics for view-dependent color, achieving a balance between compactness and expressiveness. However, the trade-off between the regularization benefits of implicit fields and the raw speed of pure explicit optimization remains an active area of investigation. A critical open question lies in determining the optimal density of implicit supervision required to stabilize sparse-view optimization without sacrificing the real-time performance that defines the 3DGS paradigm. Future research must explore adaptive mechanisms that dynamically adjust the influence of implicit regularizers based on view density and scene complexity, potentially unlocking a new class of hybrid models that are both data-efficient and computationally lean. The consensus across [26] and [15] suggests that the future of high-fidelity reconstruction lies in these hybrid architectures, where implicit regularization serves as a foundational mechanism to stabilize optimization, enhance generalization in data-scarce regimes, and unlock representational capacities beyond the reach of independent Gaussian primitives.

Expressive Texture and Appearance Modeling

Standard 3DGS primitives, while revolutionary for real-time rendering, often struggle to capture high-frequency geometric details and complex specular reflections due to their reliance on simplified kernel assumptions and low-order spherical harmonics for view-dependent color. To overcome these representational bottlenecks, recent research has pivoted toward augmenting Gaussian primitives with neural texture fields, thereby shifting from static attribute storage to dynamic, learned appearance modeling. A primary limitation of early enhancements involving per-splat 2D texture maps is their tendency to lack global consistency, leading to redundancy and overfitting, particularly in sparse-view scenarios. In contrast, neural texture approaches utilize shared global representations to ensure robustness across varying view densities. [3] introduces a framework that addresses these deficiencies by augmenting each Gaussian primitive with local RGBA texture fields predicted by a global neural field, represented as a hybrid of a tri-plane and a neural decoder. This architecture allows for the efficient modeling of view- and time-dependent effects, which are critical for capturing specularities in static scenes and temporal appearance variations in dynamic reconstructions. By leveraging implicit regularization through a shared global network, this method enforces spatial coherence across primitives, significantly improving generalization in sparse-view settings where independent per-primitive optimization typically fails.

The challenge of maintaining expressive appearance while managing computational and storage costs has led to divergent strategies in the literature. While [3] emphasizes increasing representational capacity through global neural fields, other works focus on compactness without sacrificing fidelity. [184] proposes replacing standard spherical harmonics with a grid-based neural field to represent view-dependent color more compactly, alongside learnable mask strategies and codebooks to reduce the total number of Gaussians. This highlights a tension in the field between expanding the expressive power of individual primitives via complex neural decoders and compressing the overall representation to facilitate deployment. Similarly, [187] argues that single-color primitives are non-compact for complex scenes, proposing the use of spatially varying functions, including tiny neural networks within primitives, to model local texture variations directly. This approach aligns with the broader goal of enhancing the intrinsic expressiveness of the primitive itself rather than relying solely on external feature grids, suggesting that local complexity can sometimes be more efficient than global over-parameterization.

In dynamic scenes, the requirement for expressive texture modeling becomes even more acute due to the need to disentangle motion from appearance changes. [30] advances this by replacing spherical harmonics with neural features that facilitate the modeling of view- and time-dependent appearance while maintaining a small storage footprint. This method formulates expressive Spacetime Gaussians enhanced with temporal opacity and parametric motion, allowing the capture of transient content and complex lighting changes over time. Furthermore, [40] specifically targets the difficulty of representing specular surfaces in dynamic environments by combining 3DGS with physically-based rendering and deformation fields. It introduces a residual correction technique for surface normals and a deformable environment map, demonstrating that standard neural textures may still fall short in physically accurate specular reconstruction without explicit physical priors. This indicates a critical divergence: while neural fields offer flexibility, they may require hybridization with physical models to achieve true material fidelity in dynamic settings.

The integration of neural textures also plays a pivotal role in handling unconstrained or "in-the-wild" conditions where illumination and occlusions vary significantly. [96] leverages robust DINO features integrated within an appearance modeling module to handle such variations, surpassing baselines that struggle with explicit representations in uncontrolled settings. This suggests that neural texture fields can serve as a vessel for injecting high-level semantic or robust visual features into the rendering pipeline, moving beyond low-level color prediction. Additionally, for scenarios involving extreme lighting conditions, [36] and [294] extend the color dimensionality of Gaussians to handle high dynamic range data, utilizing asymmetric grids and linear HDR spaces to prevent the baking of exposure discrepancies into the radiance field. These methods underscore that expressive appearance modeling is not merely about texture resolution but also about the physical correctness of the radiance representation, challenging the assumption that LDR training data is sufficient for high-fidelity reconstruction.

Further refinements in texture modeling address specific domain constraints, such as underwater imaging and surgical scenes. [196] and [211] incorporate learnable underwater image formation modules to model wavelength-dependent attenuation and backscatter, effectively treating the water medium as a learnable texture component. In the medical

domain, [39] introduces spatially adaptive color encoding using dynamic anchors to capture subtle tissue color variations that fixed per-Gaussian colors miss. This approach mirrors the logic of [3] by moving away from static attributes toward locally adaptive, learned color fields. Finally, [2] reinforces the necessity of global regularization, modeling splat features as outputs of an implicit neural field to introduce spatial autocorrelation, which is essential for stabilizing optimization when input views are sparse. Collectively, these works illustrate a clear trajectory toward hybrid representations that combine the efficiency of explicit Gaussian primitives with the expressive, regularized power of neural texture fields to achieve high-fidelity reconstruction across diverse and challenging scenarios. Future research must address the remaining gap in fully disentangling intrinsic material properties from dynamic lighting in uncontrolled environments, as current methods often rely on scene-specific training that limits generalization to novel illumination conditions.

Handling Sparse Views and Generalization

The dependency of 3DGS on dense, calibrated multi-view inputs constitutes a primary bottleneck for its deployment in practical, unconstrained environments. Without sufficient geometric constraints from multiple viewpoints, standard splat-based techniques are highly susceptible to overfitting training views, resulting in floaters, geometric inconsistencies, and poor generalization to novel perspectives. This vulnerability stems from the independent optimization of Gaussian primitives, which lacks the inherent spatial bias found in implicit neural representations. To mitigate these issues, recent research has pivoted toward hybridizing explicit Gaussian frameworks with implicit neural fields and structural regularizers. A seminal approach in this direction is presented in [2], which identifies the lack of spatial autocorrelation among splat features as a critical failure mode in sparse settings. The authors propose regularizing splat behavior by modeling features as outputs of a corresponding implicit neural field, utilizing hierarchical convolutional decoders to enforce spatial bias. This method effectively bridges the gap between the efficiency of explicit splatting and the robustness of implicit models, demonstrating that introducing such constraints is essential for high-quality reconstruction from limited inputs. Similarly, [3] augments Gaussian primitives with local texture fields predicted by a global tri-plane network, enforcing global consistency across primitives to improve robustness in both sparse and dense settings. These hybrid approaches suggest that the raw explicit representation of 3DGS requires augmentation through neural fields to generalize effectively beyond laboratory settings.

Complementing the use of implicit fields, other works have explored self-augmentation strategies to synthetically increase view density or leverage external priors to compensate for missing geometric information. [275] addresses the scarcity of input data by generating a basic 3D representation from sparse views and then employing a pre-trained 2D diffusion model to enhance rendered multi-view images. These augmented images serve as additional supervision signals to refine the 3D Gaussians, while a structural masking strategy enhances robustness against noise. This two-stage framework highlights a divergent yet effective pathway: rather than altering the internal representation of the Gaussians, one can enrich the training signal itself. In scenarios where even synthetic augmentation is insufficient due to extreme sparsity or lack of camera poses, methods like [98] model continuous volumetric density through inner 3D Gaussian distributions to reconstruct internal structures from sparse sliced data without requiring camera poses. Furthermore, [179] leverages the dense point clouds generated by 3DGS to guide neural implicit surface reconstruction, introducing a scale regularizer to force Gaussian centers onto surfaces and refining point clouds with normal priors to achieve detailed geometry even when direct supervision is sparse.

The challenge of sparse views is further compounded in dynamic scenarios and monocular settings, where temporal consistency must be maintained alongside spatial generalization. [93] tackles the scarcity of training data for dynamic scenes by introducing a feed-forward method that predicts deformable 3D Gaussians from monocular videos. By leveraging a large-scale synthetic dataset with dense 3D scene flow supervision, this model learns a generalizable prior that reduces the dependency on dense multi-view captures for any specific scene. In a similar vein, [130] integrates deformable Gaussian splatting with dynamic neural surfaces, where depth maps from the splatting module guide ray sampling and provide supervision for the neural surface, while the neural surface directs the distribution of Gaussian primitives. This reciprocal relationship ensures that geometric constraints are maintained even with limited monocular input. Additionally, [232] enhances the priors driving dynamic reconstruction by refining object-level masks and depth estimates from foundation models, demonstrating that upgrading the quality of 2D priors can significantly mitigate the ill-posed nature of monocular dynamic reconstruction.

Generalization also extends to handling unconstrained capture conditions, such as varying exposures, motion blur, and unknown camera poses, which frequently accompany casual data collection. [32] proposes a unified differentiable physical imaging model to reconstruct HDR scenes from casually captured videos with unknown exposure times. By jointly optimizing exposure, camera response functions, and scene geometry, this method demonstrates robustness where traditional methods fail due to inconsistent lighting. Furthermore, [202] addresses the accumulation of pose errors in monocular dynamic reconstruction by incorporating camera poses as learnable parameters within a unified optimization framework. This end-to-end approach mitigates the impact of motion blur on pose estimation, thereby improving reconstruction quality in scenarios where precise camera tracking is unavailable. For crowd-sourced imagery characterized by extreme sparsity and pose uncertainty, [153] utilizes robust geometric priors and advanced illumination modeling to handle these irregularities, while [96] leverages robust DINO features and appearance

modeling modules to handle occlusions and illumination changes that typically degrade explicit representations in uncontrolled environments.

Despite these advances, significant open questions remain regarding the scalability and computational trade-offs of these hybrid approaches. While methods like [2] and [3] successfully introduce implicit biases, they often incur higher training costs or require complex architectural modifications that may hinder real-time applications. Moreover, the reliance on pre-trained foundation models for priors, as seen in [275] and [232], raises concerns about the generalization limits when encountering domains significantly divergent from the pre-training data. Future research must address how to distill these powerful priors into lightweight, scene-specific regularizers that maintain the real-time rendering advantages of 3DGS without sacrificing the robustness gained from hybridization. Additionally, there is a need for unified frameworks that can simultaneously handle sparsity, dynamic motion, and varying illumination without requiring distinct pipeline stages or specialized loss functions for each condition, moving towards a truly generalizable solution for 3D reconstruction in the wild. The evolution from purely optimization-based splatting to these hybrid, prior-enhanced frameworks marks a critical transition in making 3D reconstruction viable for real-world applications where dense, calibrated multi-view data is rarely available.

Optimization Efficiency and Temporal Consistency

Achieving real-time performance in dynamic reconstruction necessitates a fundamental shift from treating all scene elements as equally dynamic to adopting strategies that explicitly differentiate between static and moving components. A primary source of computational inefficiency in early dynamic 3DGS methods was the redundant modeling of static backgrounds with time-varying parameters, leading to excessive memory consumption and prolonged training times. To mitigate this, recent frameworks have prioritized dynamic-static decomposition. [25] addresses this by utilizing a feed-forward encoder and optical flow model to eliminate redundant computations on static regions, reportedly reducing training time to minutes while maintaining high rendering speeds. Similarly, [152] employs motion-related Gaussians exclusively for dynamic components while retaining vanilla Gaussians for static areas, a strategy that effectively prevents flickering artifacts and improves storage efficiency. This separation is further refined in [20], which adaptively converts temporally invariant Gaussians into a compact 3D representation, reserving full 4D representations only for dynamic elements to optimize the trade-off between fidelity and computational cost. These approaches collectively demonstrate that identifying and isolating static regions is not merely an optimization trick but a structural necessity for scalable dynamic reconstruction.

The literature reveals a significant divergence in how motion is parameterized to ensure both efficiency and consistency, creating a tension between model compactness and rendering speed. [147] highlights that while structure-guided methods often achieve superior reconstruction fidelity and compactness, Gaussian-centric approaches offer higher rendering speeds but suffer from quality variability and storage overhead. To resolve this, some methods argue for structured representations to reduce optimization complexity. [112] posits that complex motions can be decomposed into a small set of learned neural trajectories, binding points to shared motion coefficients to significantly reduce storage. In contrast, [199] reinterprets the problem as a continuous-time dynamical system, integrating a learned neural dynamical field to achieve better motion coherence and temporal extrapolation capabilities compared to discrete deformation baselines. Meanwhile, [242] introduces a hierarchical coarse-to-fine pipeline and rate-distortion optimization to reduce deformation redundancy, claiming substantial improvements in rendering speed over per-Gaussian approaches. This divergence suggests that the optimal parameterization depends heavily on the specific constraints of the application, whether it be storage-limited mobile deployment or high-fidelity offline rendering.

Temporal consistency, particularly in streaming applications, is often compromised by error accumulation and the inability to model long-term dependencies without prohibitive costs. Sliding window approaches have emerged as a critical solution to address the high cost of long-sequence training and to maintain local temporal coherence. [23] introduces an adaptive sliding window strategy that partitions sequences into manageable segments, allowing the canonical representation to evolve and handling significant geometric changes while enforcing consistency through overlapping frames. This concept is echoed in [124], which leverages a temporal Transformer within a short window of frames to model motion dependencies, preventing correspondence drift under large motions. Furthermore, [154] utilizes a bidirectional deformation field to recover scene dynamics from temporally local observations, ensuring stability in online reconstruction of arbitrarily long video streams without requiring calibrated inputs. These methods illustrate a move away from global optimization towards localized, incremental updates that balance accuracy with computational feasibility.

To address inconsistent observations and flickering in static regions, error compensation mechanisms and adaptive sampling strategies are essential. [97] proposes a key-frame-guided streaming strategy that refines specific frames to mitigate error accumulation, achieving faster per-frame reconstruction times. Complementing this, [91] introduces a perturbation smoothing strategy and a hierarchical coherent motion mechanism that merges additional Gaussians into an initial representation, removing low-opacity primitives to maintain a compact and consistent state. Adaptive sampling is also central to [65], which utilizes a selective inheritance stage to preserve temporal continuity and an error-guided densification stage to accommodate emerging objects, distinguishing between dynamic and static primitives to optimize their movements efficiently. Despite these advancements, contradictions remain regarding the optimal balance between model compactness and rendering speed, suggesting that future research must focus on unified frameworks that can dynamically adapt their representation strategy based on scene complexity and available computational resources. The integration of physical priors and external data modalities also plays a role in enhancing consistency; [204] and [64] demonstrate that incorporating event streams and physical constraints can resolve

ambiguities in fast motion and improve temporal resolution, offering a pathway to more robust dynamic reconstruction in challenging environments where visual data alone is insufficient.

Conclusion and Future Directions

The literature on dynamic 3D Gaussian Splatting reveals a decisive evolution from purely implicit neural representations toward explicit, hybrid, and structurally aware frameworks. This transition is driven by the critical necessity to reconcile conflicting demands: real-time rendering speeds, high-fidelity synthesis, and memory efficiency in complex, non-rigid environments. While early approaches relied on global deformation fields to warp canonical Gaussians, the field has matured to recognize the limitations of unified models in handling diverse motion patterns and large-scale scene changes. Consequently, contemporary research increasingly favors adaptive strategies that disentangle static and dynamic components, employ motion-aware partitioning, and utilize sparse control mechanisms to allocate computational resources precisely where needed. These methodological innovations have successfully mitigated the blurring artifacts and temporal inconsistencies that plagued earlier iterations, establishing a new standard for dynamic scene reconstruction.

A central theme emerging from this synthesis is the persistent tension between architectural compactness and representational expressiveness. Structure-guided methods, which leverage auxiliary representations such as deformation fields, anchor grids, or explicit meshes, consistently demonstrate superior geometric fidelity and model compactness. In contrast, Gaussian-centric approaches that encode dynamics directly into primitives offer unparalleled rendering speeds but often suffer from quality variability and significant storage overhead. To bridge this gap, the field is converging on hybrid architectures that integrate the regularization benefits of implicit neural fields with the efficiency of explicit splatting. By enforcing spatial autocorrelation through neural textures or tri-plane representations, researchers have stabilized optimization in sparse-view settings and enhanced the modeling of high-frequency details, effectively combining the robustness of volumetric rendering with the performance of rasterization-based techniques.

Furthermore, the extension of dynamic Gaussian Splatting into specialized domains underscores the necessity of incorporating domain-specific priors and multi-sensor fusion. Applications in autonomous driving, robotic surgery, and high-speed imaging have exposed the inadequacy of generic reconstruction pipelines when faced with challenges such as motion blur, occlusion, and extreme lighting conditions. The integration of optical flow, event camera data, depth sensors, and physical constraints has proven essential for resolving ambiguities inherent in monocular inputs and ensuring physically plausible motion trajectories. Whether through mesh-adsorbed representations that enable precise editing and simulation or physics-informed models that enforce constitutive laws, these advancements highlight a move toward resilient systems capable of inferring complex 4D structures from imperfect, heterogeneous data streams.

Looking forward, the trajectory of dynamic scene representation points toward increasingly unified and intelligent frameworks, yet several critical open questions remain. First, the field lacks robust mechanisms for automatically determining the optimal granularity of motion representation—such as the number of basis trajectories or control points—based on scene complexity, currently relying on manual tuning or heuristics. Second, handling irreversible topological changes, such as tissue cutting in surgery or object breaking, remains a significant challenge for methods reliant on fixed-topology meshes or continuous deformation fields. Third, there is a need for unified frameworks that can dynamically switch between global factorization for static periods and local sliding windows for high-dynamic events without introducing boundary artifacts. Finally, the generalization of physical priors to unseen object categories and materials requires further investigation to move beyond scene-specific optimization. As the field continues to refine motion factorization, error compensation mechanisms for online streaming, and geometric regularization techniques, dynamic 3D Gaussian Splatting is poised to become the foundational technology for next-generation applications in virtual reality, digital twins, and autonomous systems, finally delivering on the promise of real-time, high-fidelity 4D reconstruction.